

Your research topic is how to improve first year programming courses at universities.

Describe four different studies around your research topic based on the following four research strategies (methods):

- Experiment

An experiment for improving first year programming could be to check the effect of the course language on students' performance in a first-year programming course. Two groups of students are taught using identical course materials, assignments, criteria and final exam, only difference is that one group receives the instructions in the students' mother tongue, while the other group receives the course in English.

The hypothesis is "teaching introduction course in programming in the mother tongue of the student will improve the student's comprehension", particularly for students who struggle both with programming and English. The idea is that being taught will reduce the language barrier and may make it easier for the student to understand concepts, documentation and instructions if the course is taught in their mother tongue. It is also possible that a course taught in English at an early stage might benefit students as it is the standard language used in programming, both in the programming language itself both also in documentation and forums.

The students would be put in to groups randomly, but removing any students who have prior experience with programming from the sample. Additionally, the sample should exclude students who have previously studied/worked in an English-speaking environment, such as an international school.

- Case Study

An exploratory case study for the effect of the language of the instructions on students' performance in a first-year programming course, could be the same setup as the experiment had where there is a introduction course in programming taught in two languages, but for the case study we would not focus on the quantitative outcomes, instead the focus of the case study would emphasize on qualitative observations and student experience.

Throughout the course, student interviews would be done at multiple stages to track how students adapt the course language, deal with programming challenges and understand the technical terminology. The interviews would give an insight on the ways the language affects, participation, motivations that may have been overlooked during the experiment.

- Survey

The survey would be done both before and after the course. The pre-course survey would assess the student's confidence, familiarity to the language/ language proficiency. This would give insight/data about the level and experience before the course. Also questions about their habits/life situation would be on the survey as they can potentially affect the student's ability to progress in the course both a negative or a positive way.

In the post-course survey, the students would be asked the same questions again both also be able to evaluate the course in general so how satisfied they were with the course and how well they understood the course material. By comparing the responses from the pre- and post surveys, we would be able to assess whether or not the course language had an impact on the students' performance or if other variables impact a student's performance more than just the change in the language that the course is taught in.

- Action Research

Action research could be that the instructor assigns senior students who have an understanding of the course content to act as mentors for the first-year students. This could for example be that the senior students are assigned to a small group of small students where they have weekly study sessions which are compulsory to attend, where the senior student helps the first-year students with the assignments and answers questions if so needed. The goal with the action research is to improve the first-year students' confidence (help students feel more comfortable to ask for help, notice other students are the same) and improve their understanding of programming concepts.

During the course, the instructor could collect data from both student and mentor feedback and measure the performance change from earlier years of the course. Based on the feedback and performance evaluation be able to refine the methods/approach used to teach first-year students the fundamentals of programming.

Write 2-4 paragraphs per study. Show that you understand the four research strategies .